THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A pure substance that cannot be broken down into simpler substances by chemical methods is a(n): 1

(a) compound **(b)** solution **(c)** mixture **(d)** element

2. The property of metals by which they can be beaten into thin sheets is: 1

(a) brittleness **(b)** sonority **(c)** malleability **(d)** ductility

3. A mixture is: 1

(a) a pure substance
(b) a combination of two or more substances not chemically combined
(c) a single compound
(d) always a liquid

4. Which of the following is an element? 1

(a) $NaCl$ **(b)** H_2O **(c)** O_2 **(d)** CO_2

5. The property of metals by which they can be drawn into wires is: 1

(a) solubility **(b)** ductility **(c)** malleability **(d)** brittleness

6. In a compound, the elements are combined: 1

(a) without any change of properties **(b)** in any ratio
(c) in a fixed ratio by mass **(d)** physically only

7. Which of the following is a compound? 1

(a) O_3 **(b)** N_2 **(c)** Fe **(d)** CO_2

8. Metals produce a ringing sound when struck. This property is called: 1

(a) ductility **(b)** sonority **(c)** lustre **(d)** malleability

9. Air is a: 1

(a) single compound **(b)** single element **(c)** mixture of gases **(d)** pure substance

10. The symbol for sodium is: 1

(a) N **(b)** Na **(c)** S **(d)** So

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Define an element. Give two examples. 2

12. What is a metalloid? Give an example. 2

13. Why is graphite used in pencils and as electrodes? 2

14. Define a compound with one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. List three physical properties of metals with an example for each. 3

16. Explain why water is a compound and not a mixture of hydrogen and oxygen. 3

17. What is a molecule? Give two examples of molecules of elements and of compounds. 3

OR

List three physical properties of non-metals. 3

18. What are alloys? Why are they made? Give three examples. 3

OR

How do the atoms of an element differ from those of another element? Give two points. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare elements, compounds and mixtures with examples in a table of five points. 5

OR

Explain atoms and molecules with the help of H_2, O_2, H_2O and CO_2, giving the number of atoms in each. 5

20. Describe the physical properties of metals and non-metals and exceptions. 5

OR

Describe how sea water can be separated into water and salt, and what the process shows about mixtures. 5

END OF QUESTION PAPER — Total: 40 Marks

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